Examples

gobanist 0.2.0

Each example is a whole document: the code below the drawing compiles alone with `typst compile`. The examples and the SGF files they read are in the folder `examples` of the repository. They are those of pgf-go, the LaTeX package by the same author. The reference of every function and option is in the manual.

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1. A position

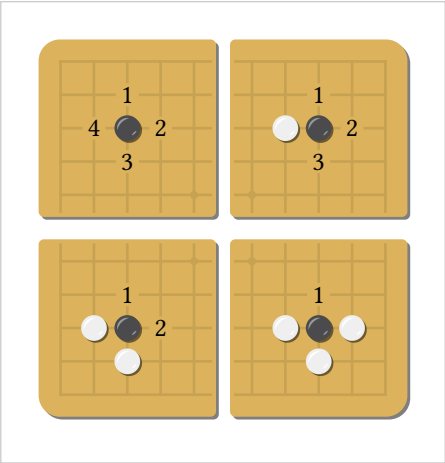



```
#import "@preview/gobanist:0.2.0": *
#set page(width: auto, height: auto, margin: 0.5cm)

#gobanist-set(profile: "pgfgosmooth")

#goban(size: 9, partial: "A2:H9",
  stones(black: "A7, B6, C8, C7"),
  stones(black: "D6, D5, D4", mark: "triangle"),
  stones(white: "C3, C4, C5, C6, D7, E7, E6"),
  stones(white: "E3", mark: (shape: "custom", content: [1])),
  marks(circle: "D3, E4, E5", line: (color: red.darken(20%))))
```

2. Liberties

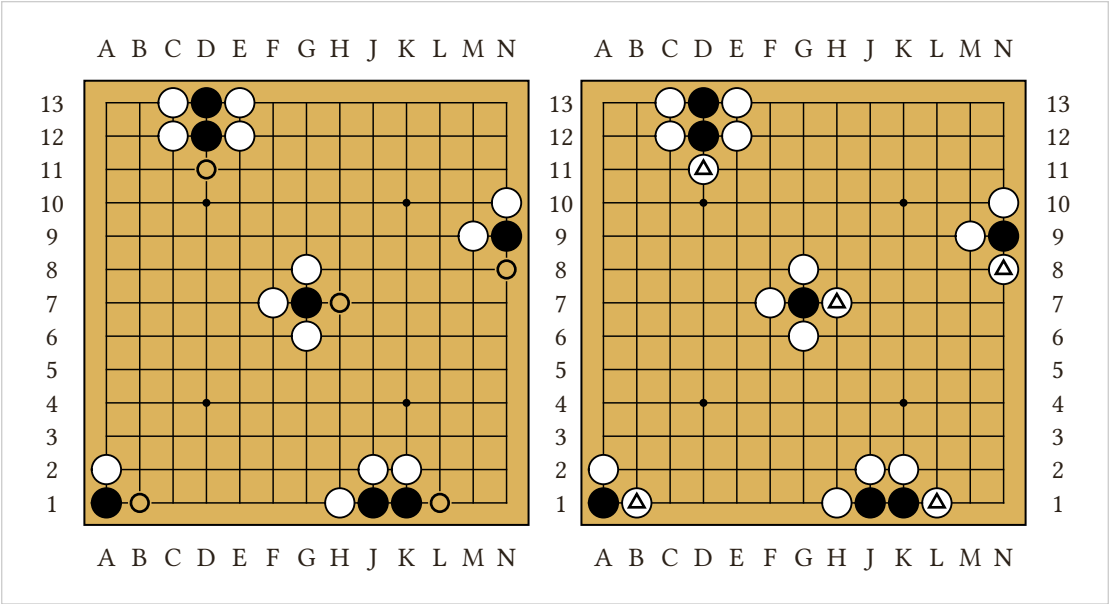



```
#import "@preview/gobanist:0.2.0": *
#set page(width: auto, height: auto, margin: 0.5cm)

#gobanist-set(profile: "pgfgosmooth")

// The liberties of a stone: four in the middle, three on the side, two in the corner.
#goban-layout(columns: 2, width: auto, column-sep: 0.5em, row-sep: 0.5em,
  board(partial: "A9:E5",
    stones(black: "C7"),
    marks(sequence: "C8, D7, C6, B7")),
  board(partial: "E9:J5",
    stones(black: "G7", white: "F7"),
    marks(sequence: "G8, H7, G6")),
  board(partial: "A1:E5",
    stones(black: "C3", white: "B3, C2"),
    marks(sequence: "C4, D3")),
  board(partial: "E1:J5",
    stones(black: "G3", white: "F3, G2, H3"),
    marks(sequence: "G4")))
```

3. Going on from a board



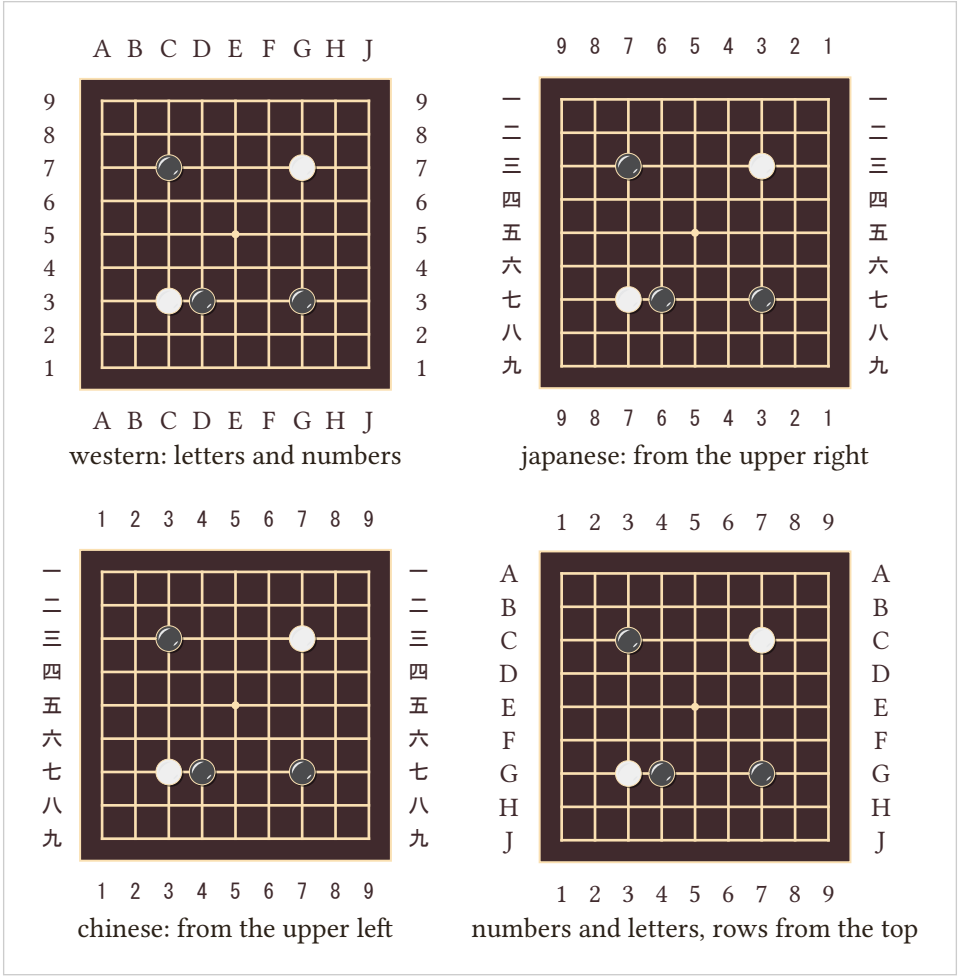

```
#import "@preview/gobanist:0.2.0": *
#set page(width: auto, height: auto, margin: 0.5cm)

#gobanist-set(background: (fill: (color: rgb(220, 179, 92))))

// A board is a value: the second one starts with the stones of the first (resume), but not with its marks.
#let atari = board(size: 13,
  stones(black: "A1, J1, K1, G7, N9, D12, D13",
    white: "A2, H1, J2, K2, G6, F7, G8, M9, N10, C12, C13, E12, E13"),
  marks(circle: "B1, L1, N8, H7, D11"))

#goban(atari, label: (at: (top, bottom, left, right)))
#goban(resume: atari, label: (at: (top, right, bottom)),
  stones(white: "B1, L1, N8, H7, D11", mark: "triangle"))
```

4. Coordinates



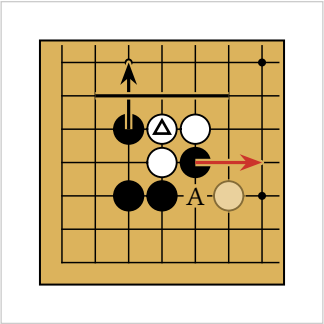

```
#import "@preview/gobanist:0.2.0": *
#set page(width: auto, height: auto, margin: 0.5cm)

// The kanji need a font that has them: MS Gothic comes with Windows; Noto Sans CJK JP is another one.
#let kanji = (font: "MS Gothic", size: 8pt)
#gobanist-set(profile: "wain", label: (at: (top, bottom, left, right)))

#let position = board(size: 9, stones(black: "C7, G3, D3", white: "G7, C3"))

#grid(columns: 2, column-gutter: 1.5em, row-gutter: 0.6em, align: center,
  goban(position),
  goban(position, label: (style: "japanese", text: kanji)),
  [western: letters and numbers], [japanese: from the upper right],
  v(0.5em), v(0.5em),
  goban(position, label: (style: "chinese", text: kanji)),
  goban(position, label: (columns: "1", rows: "A", rows-direction: ttb)),
  [chinese: from the upper left], [numbers and letters, rows from the top])
```

5. Arrows, lines and marks

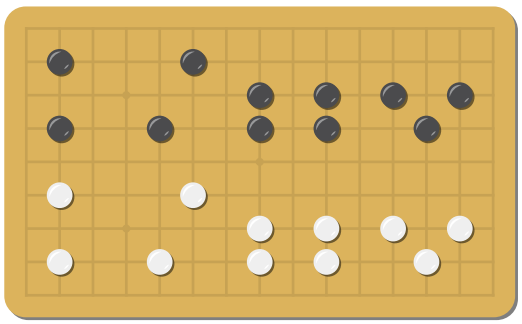



```
#import "@preview/gobanist:0.2.0": *
#set page(width: auto, height: auto, margin: 0.5cm)

#gobanist-set(background: (fill: (color: rgb(220, 179, 92))))

#goban(size: 9, partial: "A1:G7",
  stones(black: "C3, D3, E4, C5", white: "D4, D5, E5, F3"),
  marks(dim: "F3"),
  arrows(("C5", "C7")),
  arrows(("E4", "G4"), color: red.darken(20%)),
  lines(("B6", "F6")),
  marks(triangle: "D5"),
  marks(custom: "E3", content: [A]))
```

6. Shapes moved with shift



A B C D E F G H J K L M N O P

From left to right: the one-space jump, the knight's move, the bamboo joint and the tiger's mouth.

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 12cm, height: auto, margin: 0.5cm)

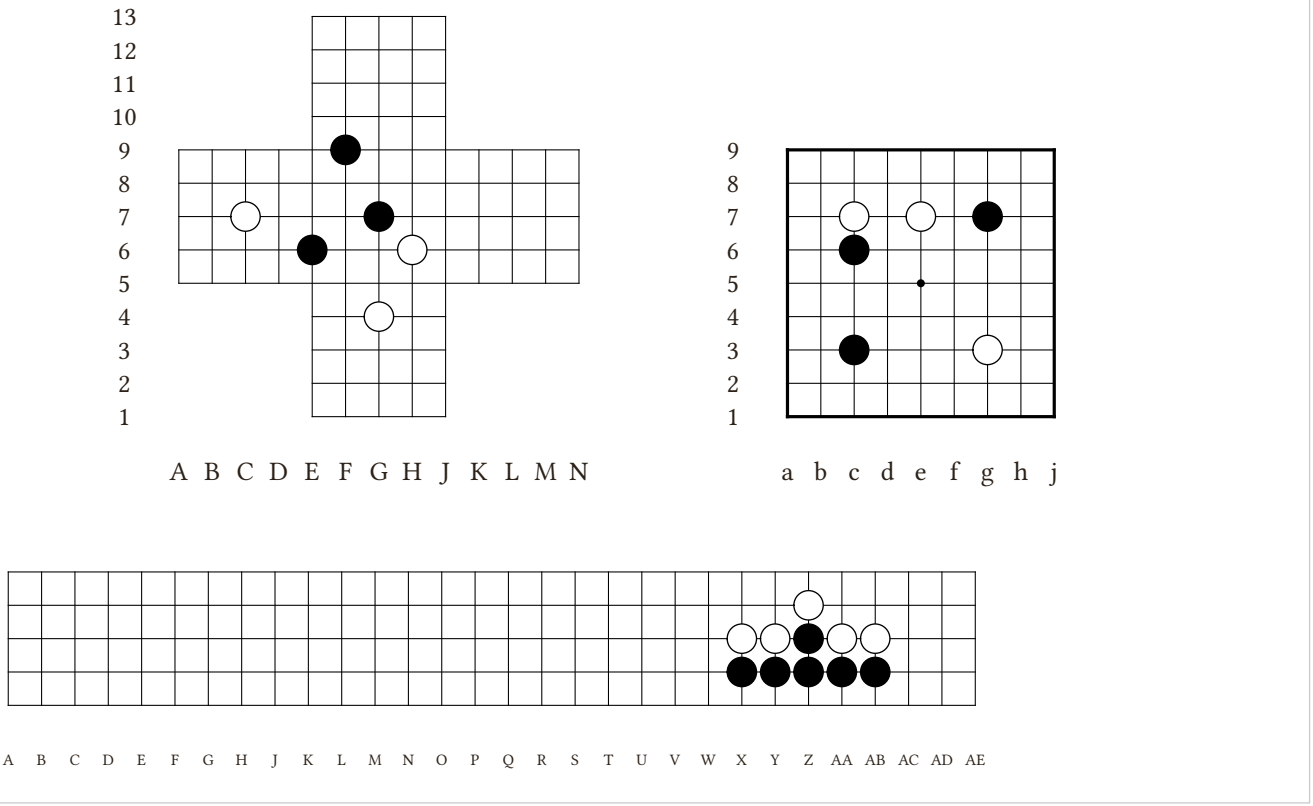
// Each shape is written once, from A1; shift: (columns, rows) draws it anywhere else.
#let jump = "A1, A3"
#let knight = "A1, B3"
#let bamboo = "A1:2, C1:2"
#let tiger = "A2, B1, C2"

#gobanist-set(profile: "pgfgosmooth")

#goban(size: (15, 9), label: (at: bottom),
  stones(black: jump, shift: (1, 5)),
  stones(black: knight, shift: (4, 5)),
  stones(black: bamboo, shift: (7, 5)),
  stones(black: tiger, shift: (11, 5)),
  stones(white: jump, shift: (1, 1)),
  stones(white: knight, shift: (4, 1)),
  stones(white: bamboo, shift: (7, 1)),
  stones(white: tiger, shift: (11, 1)))

From left to right: the one-space jump, the knight's move, the bamboo joint and the tiger's mouth.
```

7. Boards of other shapes



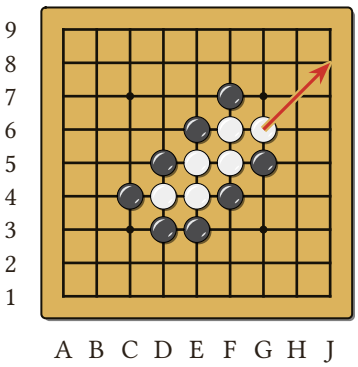

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 17cm, height: auto, margin: 0.5cm)

#gobanist-set(profile: "simple", label: (at: (bottom, left)))

// blank: points that are not part of the board (a board in the shape of a cross).
// grid.edge: the outer lines of the grid; label.columns: "a", the letters in lower case.
#goban-layout(columns: 2, width: auto, column-sep: 2em, valign: bottom,
  board(size: 13, blank: "A:D 1:4, K:N 1:4, A:D 10:13, K:N 10:13",
    stones(black: "G7, F9, E6", white: "H6, G4, C7")),
  board(size: 9, grid: (edge: (width: 1.2pt)), label: (columns: "a"),
    stones(black: "C3, G7, C6", white: "G3, E7, C7")))

// After Z, the columns have two letters: AA, AB...
#v(1em)
#goban(size: (30, 5), label: (at: bottom, text: (size: 6pt)),
  stones(black: "X2:AB2, Z3", white: "X3, Y3, AA3:AB3, Z4"))
```

8. A ladder



Black plays at G7 and White is in atari again: the chain keeps running towards the corner, where it is captured.

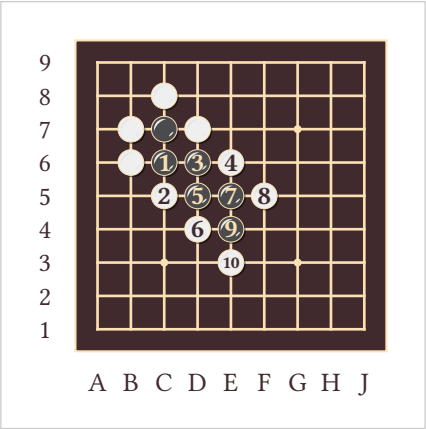
```
#import "@preview/gobanist:0.2.0": *
#set page(width: 8cm, height: auto, margin: 0.5cm)

#gobanist-set(profile: "pgfgo", label: (at: (bottom, left)))

// A ladder (shicho): the white chain is a connected staircase (-|-, then -- for its last stone) and the black
// stones that chase it are two diagonals (--). Each line is written with its ends only.
#goban(
  stones(white: "D4 -|- F6 -- G6", black: "C4, D3, D5 -- F7, E3 -- G5"),
  arrows(("G6", "J8")))

Black plays at G7 and White is in atari again: the chain keeps running towards the corner, where it is captured.
```

9. Moves

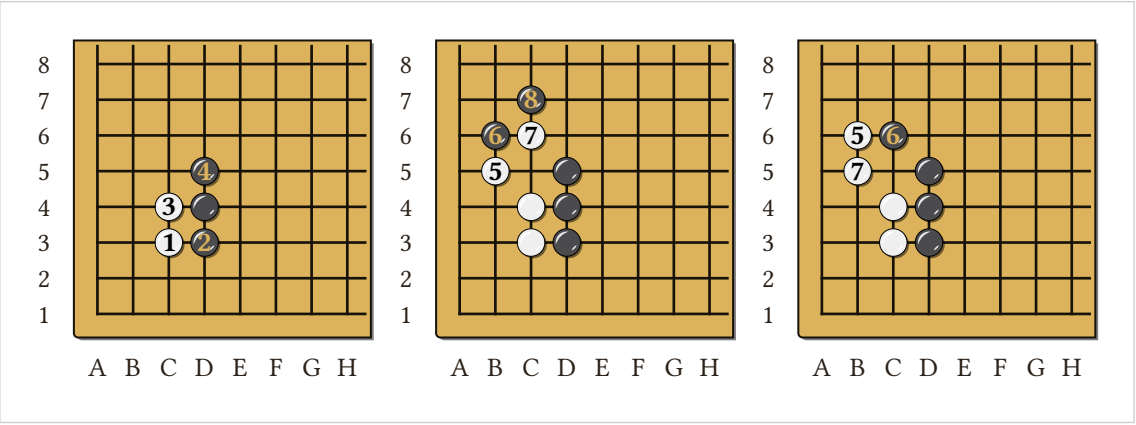



```
#import "@preview/gobanist:0.2.0": *
#set page(width: auto, height: auto, margin: 0.5cm)

#gobanist-set(profile: "wain")

#goban(size: 9, label: (at: (bottom, left)),
  stones(black: "C7", white: "B6, B7, C8, D7"),
  moves(black: "C6, C5, D6, E6, D5, D4, E5, F5, E4, E3"))
```

10. Variations




```
#import "@preview/gobanist:0.2.0": *
#set page(width: 14cm, height: auto, margin: 0.5cm)

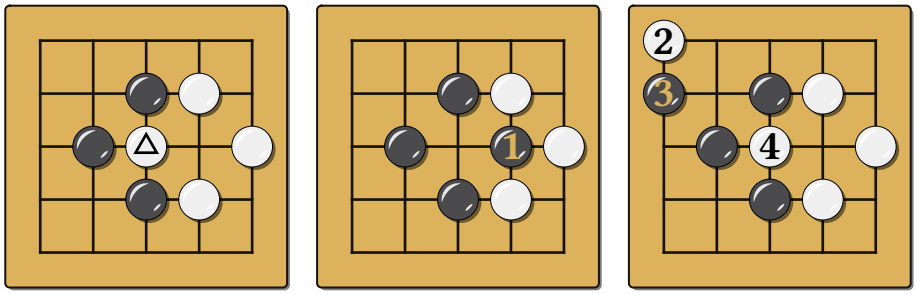
#gobanist-set(profile: "pgfgo", size: 19, partial: "A1:H8", label: (at: (bottom, left)))

// Each diagram goes on from the board of another one (resume): the stones are already there, and the
// numbers and the turn go on. The third one is a variation of the first.
#let first = board(stones(black: "D4"), moves(white: "C3, D3, C4, D5"))

#goban-layout(columns: 3, fit: true,
  first,
  board(resume: first, moves("B5, B6, C6, C7")),
  board(resume: first, moves("B6, C6, B5")))
```

11. The ko, with views

The ko



The white stone △ has three black neighbours, , and . Black plays 1 and takes it: now the shape is . White cannot take back at once, because the same position would come again. White plays a threat first, 2 , Black answers with 3 , and only then White retakes with 4 : .

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 12cm, height: auto, margin: 0.5cm)

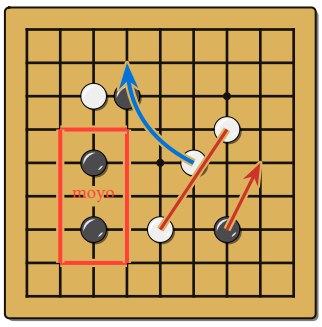
#gobanist-set(profile: "pgfgo")

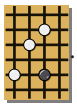
// The boards are values: the layout draws them and goban-view shows parts of them in the text.
#let ko = board(size: 5, stones(black: "B3, C2, C4", white: "C3, D2, D4, E3"))
#let before = board(ko, marks(triangle: "C3"))
#let capture = board(ko, captures: "remove", moves("D3"))
#let retake = board(ko, captures: "remove", moves("D3, A5, A4, C3"))

= The ko
#goban-layout(columns: 3, fit: true, before, capture, retake)

The white stone #goban-stone("white", mark: "triangle") has three black neighbours,
#goban-view(before, "B3, C2, C4").
Black plays 1 #goban-view(capture, "D3") and takes it: now the shape is #goban-view(capture, "B2:E4").
White cannot take back at once, because the same position would come again. White plays a threat first,
2 #goban-view(retake, "A5"), Black answers with 3 #goban-view(retake, "A4"), and only then White retakes with
4 #goban-view(retake, "C3"): #goban-view(retake, "B2:E4").
```

12. Views without arrows



White runs with  without the arrow and the line, the stones alone:  .

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 10cm, height: auto, margin: 0.5cm)

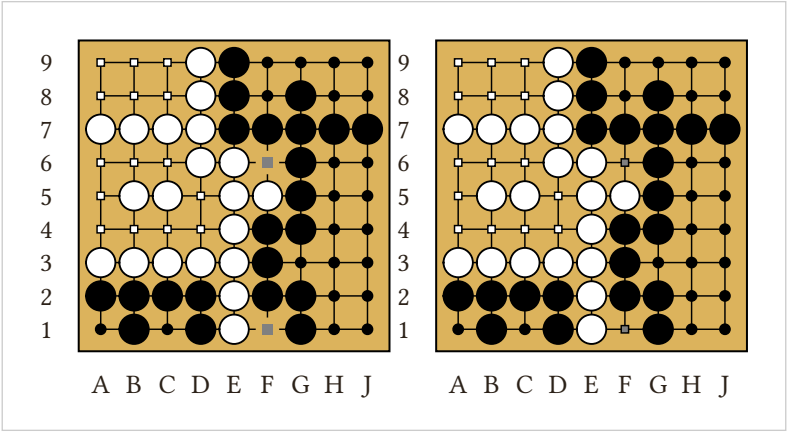
#gobanist-set(profile: "pgfgo", view: (goban: (scale: 0.45)))

// The views can leave out the arrows and the lines.
#let plan = board(size: 9,
  stones(black: "C3, C5, D7, G3", white: "E3, F5, G6, C7"),
  lines(("B2", "D2"), ("D2", "D6"), ("D6", "B6"), ("B6", "B2"), color: red, width: 1.5pt),
  marks(custom: "C4", content: text(fill: red)[moyo], label: (size: 7pt)),
  arrows(("F5", "D8"), color: blue, width: 1.5pt, bend: 30deg),
  arrows(("G3", "H5")),
  lines(("E3", "G6")))

#goban(plan)

White runs with #goban-view(plan, "E2:H7"); without the arrow and the line, the stones alone:
#goban-view(plan, "E2:H7", arrows: false, lines: false).
```

13. Territory

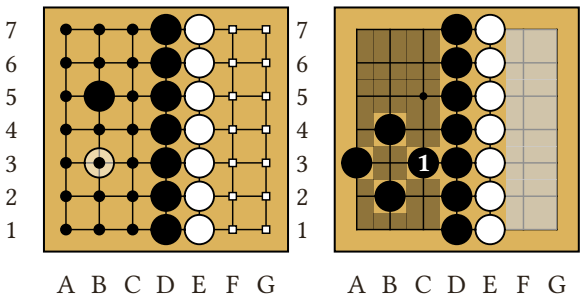



```
#import "@preview/gobanist:0.2.0": *
#set page(width: auto, height: auto, margin: 0.5cm)

#gobanist-set(background: (fill: (color: rgb(220, 179, 92))), label: (at: (bottom, left)))

// The territory marked point by point, and found by the kernel with territory(auto).
#goban(size: 9,
  stones(black: "A2, B1, B2, C2, D1, D2, G1, G2, F2, F3, F4, G4, G5, G6, E9, E8, E7, F7, G7, H7, J7, G8",
    white: "A7, B7, C7, D7, D8, D9, D6, B5, C5, A3, B3, C3, D3, E1, E2, E3, E4, E5, E6, F5"),
  marks(t-circle: "J1, J2, J3, J4, J5, J6, H1, H2, H3, H4, H5, H6, G3, H8, H9, J8, J9, G9, F9, F8, A1, C1",
    player: "black"),
  marks(t-square: "A9, A8, B9, B8, C9, C8, A6, B6, C6, A5, D5, A4, B4, C4, D4", player: "white"),
  marks(filled-square: "F1, F6", line: (color: gray)))
// The same position, with the coordinates abbreviated.
#goban(size: 9,
  stones(black: "E9|-J7, G8|-F4|-G2--G1, A2-|D1, B1",
    white: "D9|-E6--E1, F5, A:C7, A:D3, B:C5"),
  territory(auto, dame: true))
```

14. The score



Black 18, White 20.5.

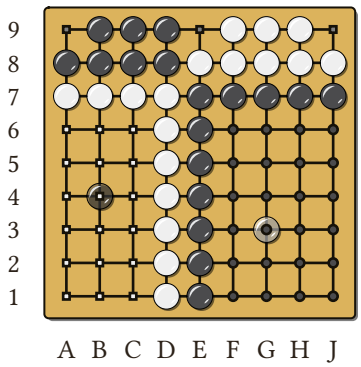
```
#import "@preview/gobanist:0.2.0": *
#set page(width: 12cm, height: auto, margin: 0.5cm)

#gobanist-set(background: (fill: (color: rgb(220, 179, 92))), label: (at: (bottom, left)))

#goban(size: 7,
  stones(black: "D1:7, B5", white: "E1:7"),
  prisoners(white: "B3"),
  territory(auto))
#goban(size: 7, captures: "remove",
  stones(black: "D1:7, A3, B2, B4", white: "E1:7, B3"),
  moves("C3"),
  territory(auto, draw: "fill"))

Black #goban-score("black"), White #goban-score("white", komi: 6.5).
```

15. A seki



Black 25, White 25.5.

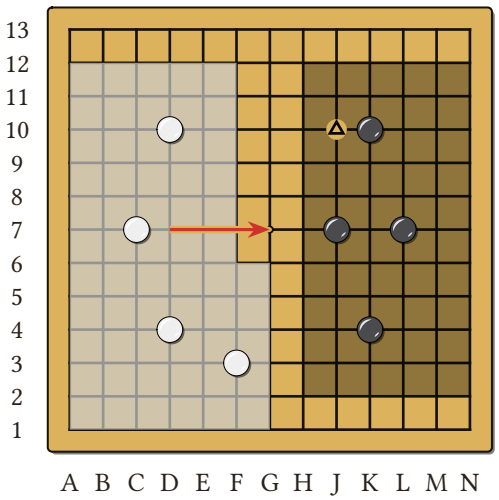
```
#import "@preview/gobanist:0.2.0": *
#set page(width: 8cm, height: auto, margin: 0.5cm)

#gobanist-set(profile: "pgfgo", label: (at: (bottom, left)))

// A seki at the top: each group has one eye (A9, J9) and they share E9. In Japanese rules the eyes of a seki
// are not territory: territory(auto) counts them, and a neutral territory for each eye takes them out.
#goban(
  stones(black: "A8:D8, B9:D9, E7:J7, E1:E6", white: "E8:J8, F9:H9, A7:D7, D1:D6"),
  prisoners(white: "G3", black: "B4"),
  territory(auto, dame: true),
  territory(neutral: "A9"),
  territory(neutral: "J9"))

Black #goban-score("black"), White #goban-score("white", komi: 6.5).
```

16. Frameworks



Black's framework: 50 squares; White's: 60. The triangle shows where White can reduce it.

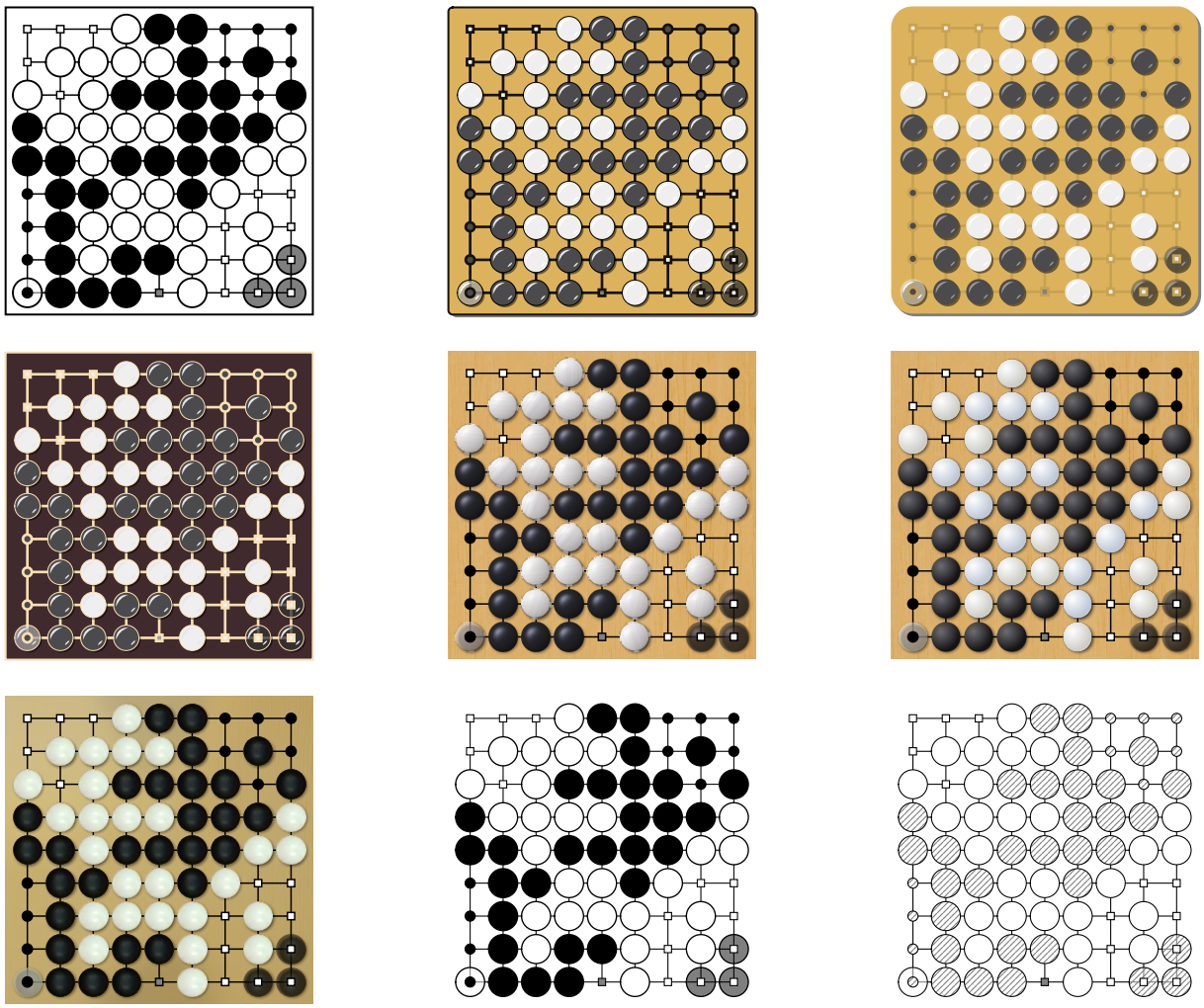
```
#import "@preview/gobanist:0.2.0": *
#set page(width: 9cm, height: auto, margin: 0.5cm)

#gobanist-set(profile: "pgfgo", size: 13, label: (at: (bottom, left)))

// Each framework is a polygon given by its corners; gobanist fills its squares and measures it.
#goban(
  stones(black: "K4, K10, J7, L7", white: "D4, D10, C7, F3"),
  territory(black: "J3, M3, M11, J11", draw: "region"),
  territory(white: "B2, F2, F5, E5, E11, B11", draw: "region"),
  marks(triangle: "J10"),
  arrows(("D7", "G7")))

Black's framework: #goban-region("black", "area") squares; White's: #goban-region("white", "area").
The triangle shows where White can reduce it.
```

17. The nine profiles

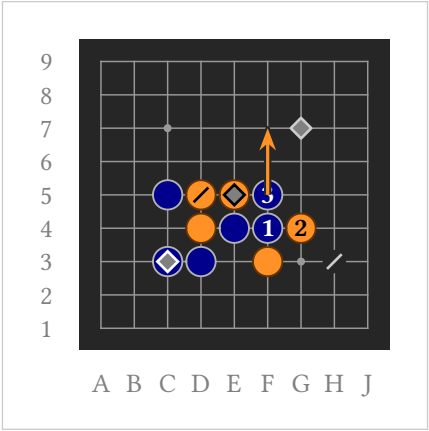


```
#import "@preview/gobanist:0.2.0": *
#set page(width: 17.4cm, height: auto, margin: 0.5cm)

// The same board with every profile: board(position, profile: p) is that board, with another profile.
#let position = board(
  stones(black: "A6|-B5|-D1|-E2, C4, E9-|F4, D5-|G7--D7, H6, H8, J7",
    white: "E8-|C5, B:E6, A7, B8, D9, J6|-H5, H2:3, F1|-C3, C2, D:E4, G4"),
  prisoners(black: "H1-|J2", white: "A1"),
  marks(t-circle: "A1:4, G8|-J9, J8, H7", player: "black"),
  marks(t-square: "G3|-J1|-H4, C9-|A8, B7", player: "white"),
  marks(t-square: "E1", fill: (color: red)))

#goban-layout(columns: 3,
  ..("default", "pgfgo", "pgfgosmooth", "wain", "shell", "yunzi", "baduktv", "simple", "draft")
    .map(p => board(position, profile: p)))
```

18. A profile of your own



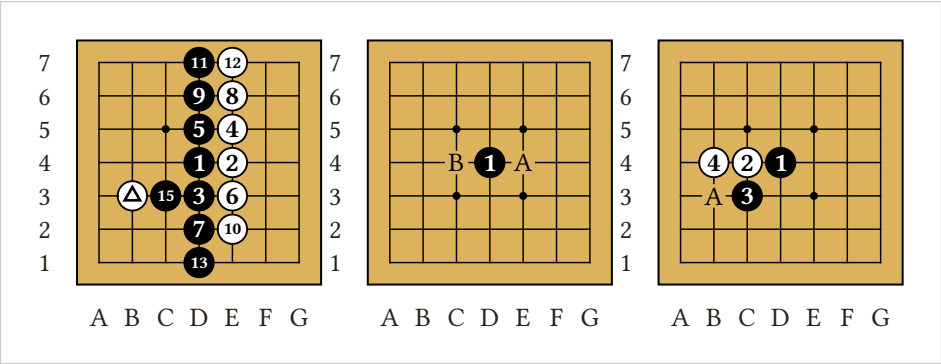

```
#import "@preview/gobanist:0.2.0": *
#set page(width: auto, height: auto, margin: 0.5cm)

// A profile of your own: a dictionary of options, like those of gobanist-set. Before setting its own,
// profile brings the style back to the initial one.
#let night = (
  background: (fill: (color: luma(15%)), line: false),
  grid: (line: (color: luma(60%))),
  label: (text: (fill: luma(50%))),
  arrow: (color: rgb(100%, 57%, 15%)),
  players: (
    neutral: (mark: (line: (color: luma(80%)), fill: (on: true, color: luma(50%)))),
    black: (stone: (fill: (color: rgb(0%, 0%, 55%)), line: (color: luma(70%)))),
    white: (stone: (fill: (color: rgb(100%, 57%, 15%)), line: (color: rgb(40%, 20%, 0%))))),
  // Marks of your own: a function of the radius of the stone and of the stroke and the fill of the mark,
  // drawn with its origin on the point.
  shapes: (
    diamond: m => polygon(fill: m.fill, stroke: m.stroke,
      (0pt, -0.7 * m.radius), (0.7 * m.radius, 0pt), (0pt, 0.7 * m.radius), (-0.7 * m.radius, 0pt)),
    slash: m => line(start: (-0.5 * m.radius, 0.5 * m.radius), end: (0.5 * m.radius, -0.5 * m.radius),
      stroke: m.stroke)),
)

#gobanist-set(profile: night)

#goban(size: 9, label: (at: (bottom, left)),
  stones(black: "C3, D3, E4, C5", white: "D4, D5, E5, F3"),
  moves("F4, G4, F5"),
  marks(diamond: "C3, E5, G7", slash: "D5, H3"),
  arrows(("F5", "F7")))
```

19. A game on a board



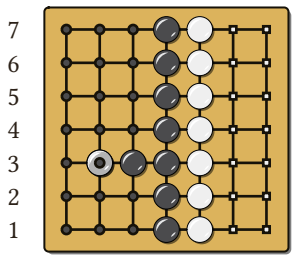

```
#import "@preview/gobanist:0.2.0": *
#set page(width: auto, height: auto, margin: 0.5cm)

#gobanist-set(background: (fill: (color: rgb(220, 179, 92))), label: (at: (bottom, left)))

// A game read from an SGF file is a value: the board takes its size, and sgf(game, ...) chooses what to draw.
#let game = sgf-read(path("commented.sgf"))

#goban(captures: "remove", sgf(game, to: 15))
#goban(sgf(game, to: 1, letters: "children"))
#goban(sgf(game, variation: (2,)))
```

20. Counting a game



Black 21, White 20.5. Result in the file: B+0.5.

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 8cm, height: auto, margin: 0.5cm)

#gobanist-set(profile: "pgfgo", label: (at: (bottom, left)))

// The end of a game: gobanist plays it with the captures, finds the territory and counts it.
#let game = sgf-read(path("commented.sgf"))

#goban(captures: "remove",
  sgf(game, marks: false, numbers: false),
  prisoners(white: "B3"), // The dead white stone.
  territory(auto))

Black #goban-score("black"), White #goban-score("white", komi: 6.5).
Result in the file: #sgf-info(game, "RE").
```

21. A commented game

A short game

Black against White, komi 6.5. A game on a 7x7 board, with comments.

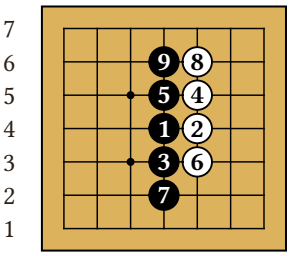


Diagram 1 (1–9)

1: Black takes the middle line.

9: Black will have three columns, White only two.

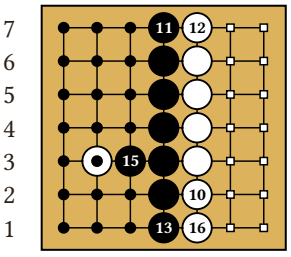


Diagram 2 (10–18)

Result: Black wins by 0.5 points (Black 21, White 20.5)

13: Tesuji. The last point of the wall.

14: Bad move. White invades, but there is no room for two eyes.

15: Black takes the vital point. The white stone is dead.

18: Black: 20 points and a prisoner. White: 14 points and 6.5 of komi.

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 13cm, height: auto, margin: 0.5cm)

#gobanist-set(background: (fill: (color: rgb(220, 179, 92))), label: (at: (bottom, left)))

#let game = sgf-read(path("commented.sgf"))

= #sgf-info(game, "GN")
#sgf-info(game, "PB") against #sgf-info(game, "PW"), komi #sgf-info(game, "KM"). #sgf-comment(game, 0)

#sgf-diagrams(game, step: 9, comments: true)
```

22. The variations of a game

A short game

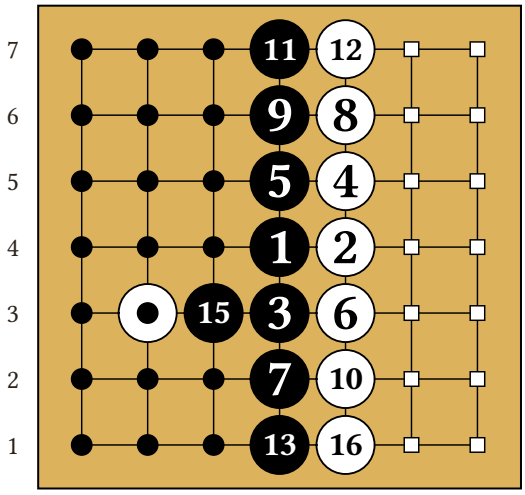


Diagram 1 (1–18)

Result: Black wins by 0.5 points (Black 21, White 20.5)

1: Black takes the middle line.
9: Black will have three columns, White only two.
13: Tesuji. The last point of the wall.
14: Bad move. White invades, but there is no room for two eyes.
15: Black takes the vital point. The white stone is dead.
18: Black: 20 points and a prisoner. White: 14 points and 6.5 of komi.

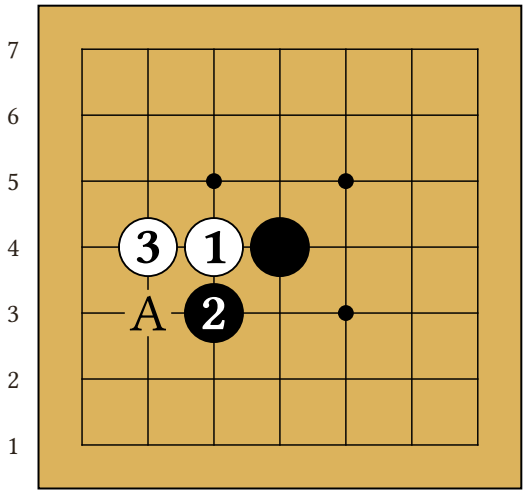


Diagram 1a (variation at 2)

1: Too close.

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 14cm, height: auto, margin: 0.5cm)

#gobanist-set(background: (fill: (color: rgb(220, 179, 92))), label: (at: (bottom, left)))

#let game = sgf-read(path("commented.sgf"))

= #sgf-info(game, "GN")

// The variations of the file get their own diagrams ("Diagram 1a"), after the diagram where they start.
#sgf-diagrams(game, step: 30, variations: (on: true, select: "all"), comments: true,
  layout: (columns: 2, fit: true))
```

23. A collection of games

Game 1: Black – White

Result: B+0.5.

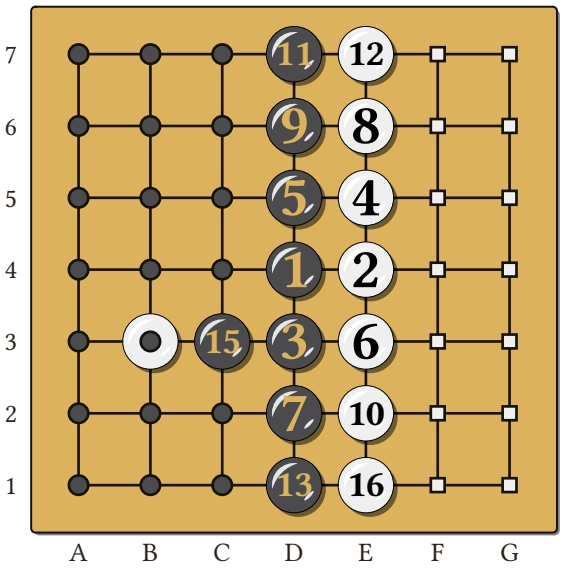


Diagram 1 (1–18)

Result: Black wins by 0.5 points (Black 21, White 20.5)

Game 2: Maeda Nobuaki – Segoe Kensaku

Result: B+R.

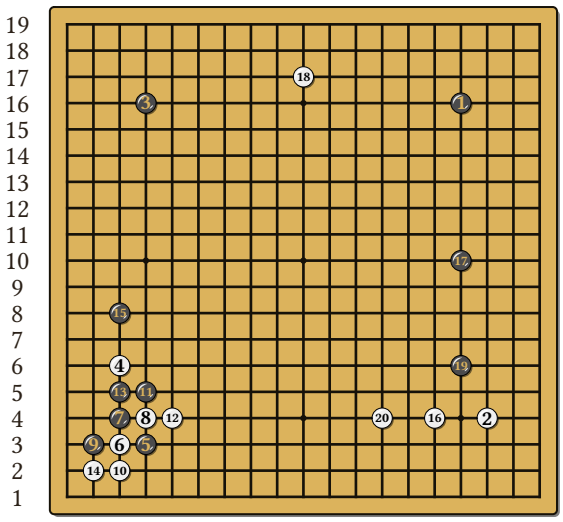


Diagram 1 (1–20)

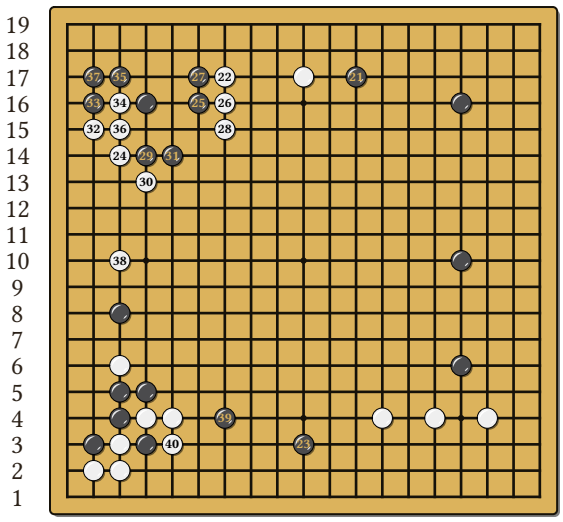


Diagram 2 (21–40)

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 15cm, height: auto, margin: 0.5cm)

#gobanist-set(profile: "pgfgo", label: (at: (bottom, left)))

// games.sgf is a collection: several games in one file, one after the other. game chooses one of them; each
// game keeps its own board size.
#for n in (1, 2) {
  let game = sgf-read(path("games.sgf"), game: n)
  [
    == Game #n: #sgf-info(game, "PB") -- #sgf-info(game, "PW")
    Result: #sgf-info(game, "RE").

    #sgf-diagrams(game, step: 20, to: 40, layout: (columns: 2, fit: true))
  ]
}
```

24. A whole game

Maeda Nobuaki – Segoe Kensaku

Black: Maeda Nobuaki 6d. White: Segoe Kensaku 7d. Komi 4.5. Played in 1940.

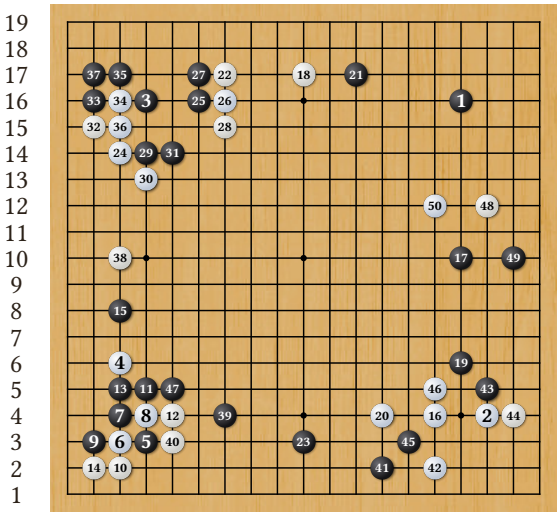


Diagram 1 (1–50)

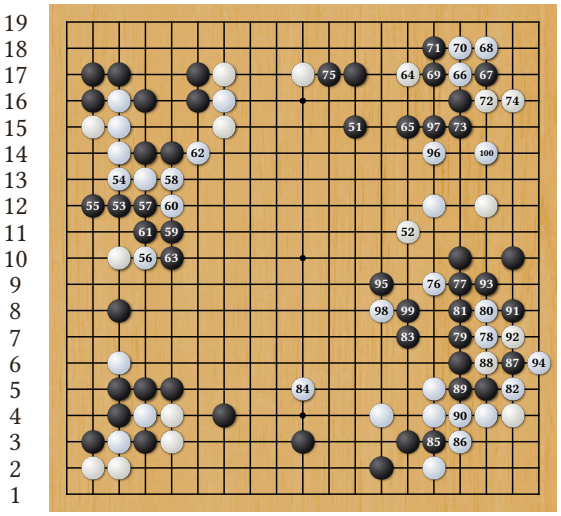


Diagram 2 (51–100)

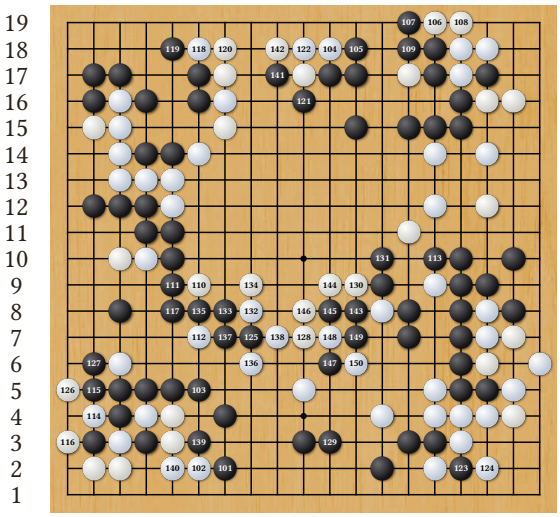


Diagram 3 (101–150)

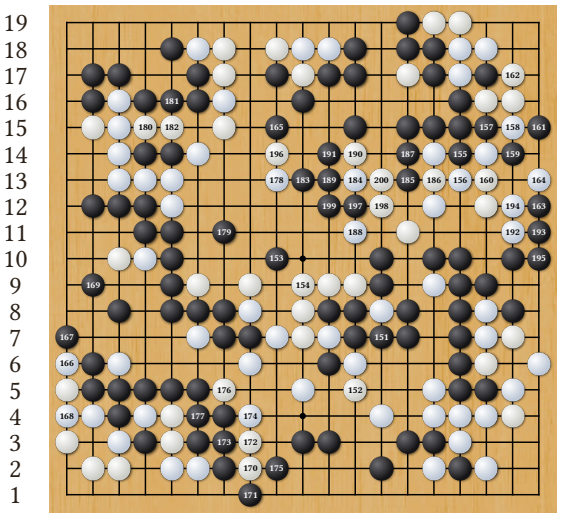


Diagram 4 (151–200)

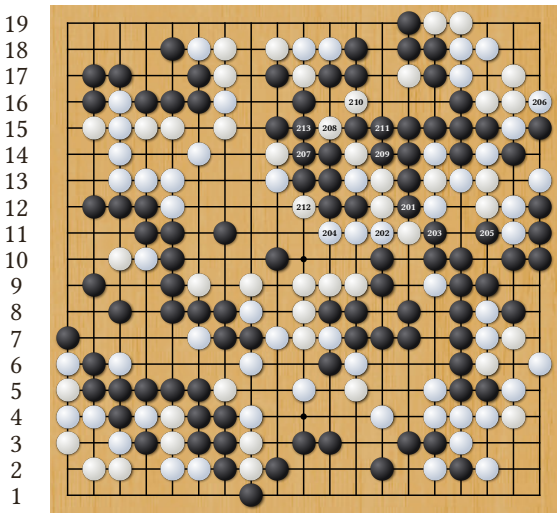


Diagram 5 (201–213)
Result: Black wins by resignation

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 15cm, height: auto, margin: 0.5cm)

#gobanist-set(profile: "yunzi", label: (at: (bottom, left)))

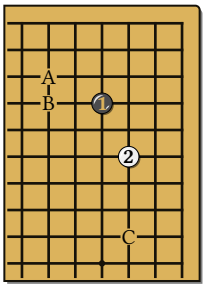
// maeda-segoe.sgf: Maeda Nobuaki against Segoe Kensaku, 1940 (moves only).
#let game = sgf-read(path("maeda-segoe.sgf"))

= #sgf-info(game, "PB") -- #sgf-info(game, "PW")
Black: #sgf-info(game, "PB") #sgf-info(game, "BR").
White: #sgf-info(game, "PW") #sgf-info(game, "WR").
Komi #sgf-info(game, "KM"). Played in 1940.

#sgf-diagrams(game, step: 50, goban: (captures: "remove"), layout: (columns: 2, fit: true))
```

25. A joseki

The 4-4 point: the low approach



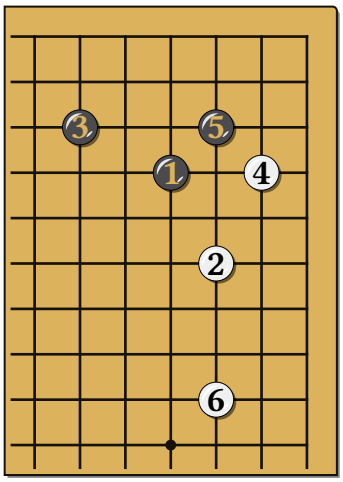


Diagram 1 (1–6)

2: The low approach.
3: A: the knight's move.
4: White slides into the corner.
6: And extends along the side.

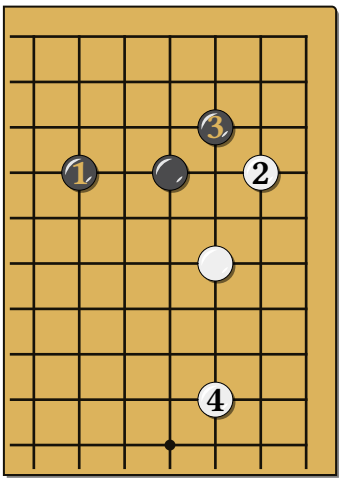


Diagram 1a (variation at 3)

1: B: the one-space jump.

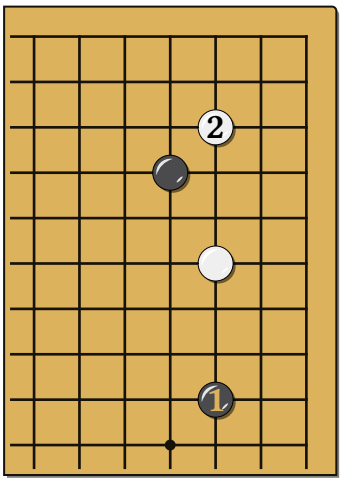


Diagram 1b (variation at 3)

1: C: the low pincer.
2: White invades at the 3-3 point.

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 14cm, height: auto, margin: 0.5cm)

#gobanist-set(profile: "pgfgo")

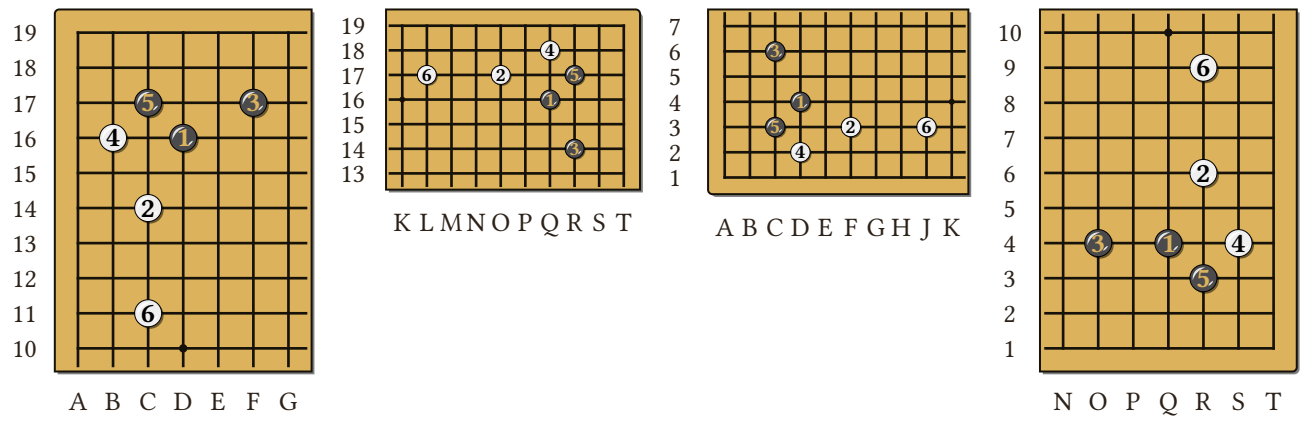
#let joseki = sgf-read(path("joseki.sgf"))

= The 4-4 point: the low approach

// The line is given by its moves, as in a joseki dictionary. The answers are the commented variations that
// leave it at move 3 or later (not the 3-3 invasion, at move 2). partial: auto shows the part of the board
// that all the diagrams use.
#align(center, goban(partial: auto, scale: 0.8, sgf(joseki, to: 2)))

#sgf-diagrams(joseki, variation: "Q16, R14", variations: (on: true, from: 3), partial: auto,
  comments: true, layout: (columns: 3, fit: true))
```

26. The symmetries of the board



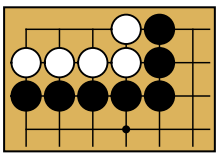

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 17cm, height: auto, margin: 0.5cm)

#gobanist-set(profile: "pgfgo", partial: auto, label: (at: (bottom, left)))

// joseki.sgf is in the upper right corner. origin says where the point aa of the file goes (initially, top
// left), and transpose swaps columns and rows (the approach comes from the other side): together, the eight
// symmetries of the board. The variation is chosen by its moves on the board.
#let file = path("joseki.sgf")

#goban-layout(columns: 4, fit: true,
  sgf-read(file, origin: top + right),
  sgf-read(file, origin: bottom + right, transpose: true),
  sgf-read(file, transpose: true),
  sgf(sgf-read(file, origin: bottom + left), variation: "Q4, R6, O4"))
```

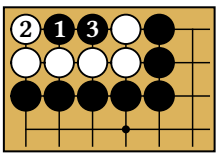
27. A problem



Problem 1:

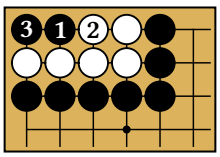
Black to play

Black to kill.



Variation 1

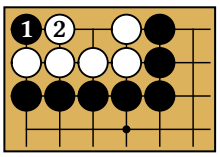
1: Tesuji. Correct: the vital point.



Variation 2

1: Tesuji. Correct: the vital point.

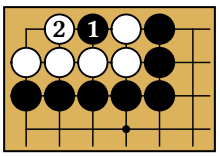
3: White has only one eye.



Variation 3

1: Bad move.

Wrong: White lives.



Variation 4

1: Wrong.

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 17cm, height: auto, margin: 0.5cm)

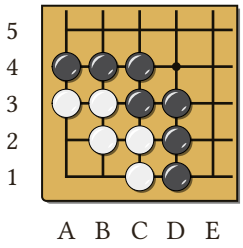
#gobanist-set(background: (fill: (color: rgb(220, 179, 92))))

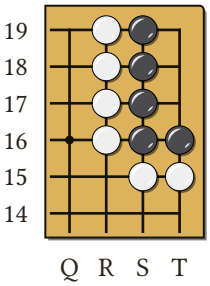
// The statement and a diagram for each line of the solution, with its comments.
#sgf-problem(sgf-read(path("problem.sgf")), title: "Problem 1")
```

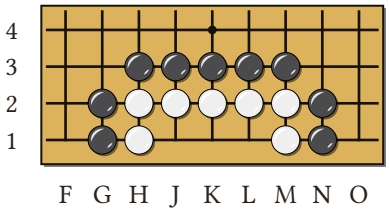
28. Problems drawn by hand

Life and death

1. Black to kill. **2.** Black to live. **3.** Black to kill.

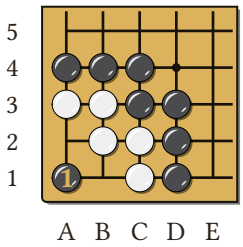


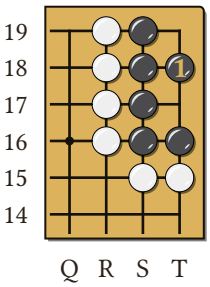


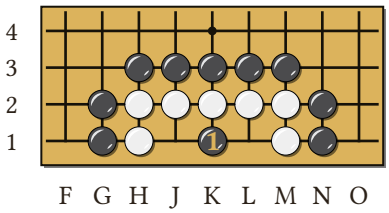


Answers

1. The eye space of White is a bent three, and its vital point is the one that touches the other two: Black plays at the corner, ①. **2.** Black lives with ①, which makes two eyes, at T17 and T19. **3.** The eye space of White is a straight three: Black plays in the middle, ①, and White has one eye at most.








```
#import "@preview/gobanist:0.2.0": *
#set page(width: 17cm, height: auto, margin: 0.5cm)

#gobanist-set(profile: "pgfgo", size: 19, partial: auto, scale: 1.1, label: (at: (bottom, left)),
  view: (goban: (background: false, label: (at: ()))))

// Problems drawn by hand, each one in another part of the board: partial: auto shows only the part that is
// used, with one more line. The answers go on from the problems, and the text shows them before they are drawn.
#let one = board(stones(white: "A:B3, B:C2, C1", black: "A:C4, C3, D1:3"))
#let two = board(stones(black: "S16:19, T16", white: "R16:19, S:T15"))
#let three = board(stones(white: "H1:2, J:L2, M1:2", black: "G1:2, H:M3, N1:2"))
#let answers = (board(one, moves("A1")), board(two, moves("T18")), board(three, moves("K1")))

= Life and death
*1.* Black to kill. #h(1em) *2.* Black to live. #h(1em) *3.* Black to kill.

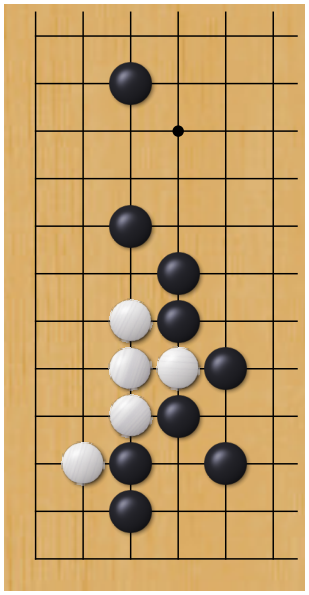
#goban-layout(columns: 3, width: auto, one, two, three)

= Answers
*1.* The eye space of White is a bent three, and its vital point is the one that touches the other two:
Black plays at the corner, #goban-view(answers.at(0), "A1").
*2.* Black lives with #goban-view(answers.at(1), "T18"), which makes two eyes, at T17 and T19.
*3.* The eye space of White is a straight three: Black plays in the middle, #goban-view(answers.at(2), "K1"),
and White has one eye at most.

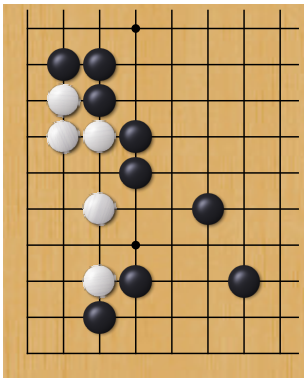
#goban-layout(columns: 3, width: auto, ..answers)
```

29. Problems from a collection

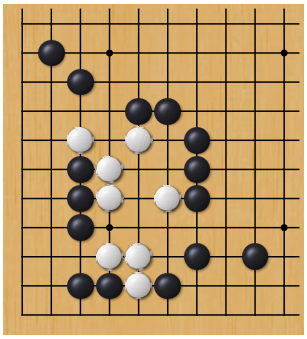
Problems from the Igo Hatsuyōron



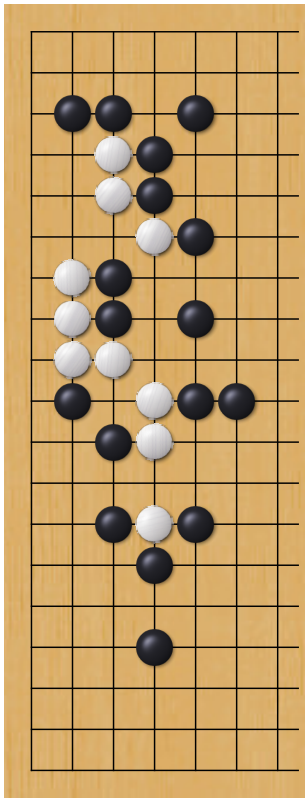
Problem 1: Black to play



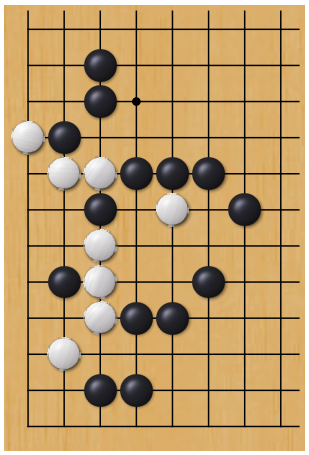
Problem 2: Black to play



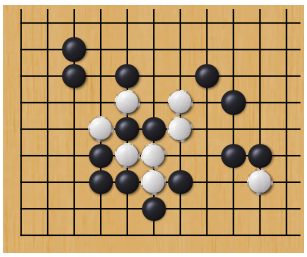
Problem 3: Black to play



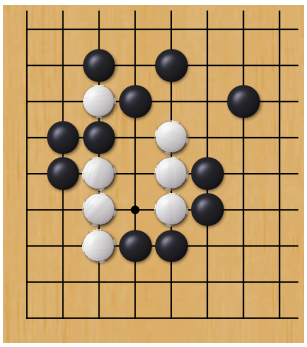
Problem 6: Black to play



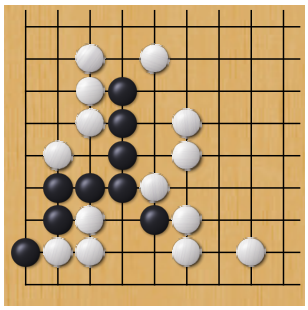
Problem 7: Black to play



Problem 9: Black to play



Problem 10: Black to play



Problem 11: Black to play

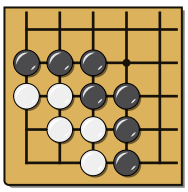
```
#import "@preview/gobanist:0.2.0": *
#set page(width: 17cm, height: auto, margin: 0.5cm)

#gobanist-set(profile: "shell")

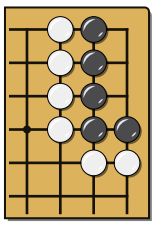
// hatsuyoron.sgf: problems of the Igo Hatsuyōron (Inoue Dōsetsu Inseki, 1713), one game per problem, with the
// position only (no solutions). The numbers of the captions are those of the file.
#let problems = (1, 2, 3, 6, 7, 9, 10, 11).map(n => sgf-read(path("hatsuyoron.sgf"), game: n))

= Problems from the Igo Hatsuyōron
#sgf-problem(..problems, to-play: "black", comments: false, layout: (columns: 4, fit: true))
```

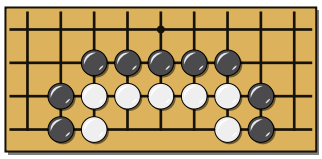
30. Problems from Sensei's Library



Problem 1: Black to play



Problem 2: Black to play



Problem 3: Black to play

````
#import "@preview/gobanist:0.2.0": *
#set page(width: 17cm, height: auto, margin: 0.5cm)

#gobanist-set(profile: "pgfgo")

// Three problems written in the document, in the syntax of Sensei's Library: one game per diagram. The boards
// have different heights; align-captions puts their captions on the same line.
#let problems = ```
$$B Black to kill
$$ | . . . . .
$$ | X X X . .
$$ | O O X X .
$$ | . O O X .
$$ | . . O X .
$$ +----------

$$B Black to live
$$ ----------+
$$ . O X . |
$$ . O X . |
$$ . O X . |
$$ . O X X |
$$ . . O O |
$$ . . . . |

$$B Black to kill
$$ . . . . . . . . .
$$ . . X X X X X . .
$$ . X O O O O O X .
$$ . X O . . . O X .
$$ ------------------
```.text

#sgf-problem(..range(1, 4).map(n => sgf-read(problems, game: n)),
  layout: (columns: 3, align-captions: true))
````
````

31. A game in Japanese

短い対局：黒対白、コミ 6.5。 七路盤の対局。

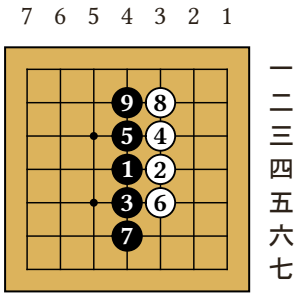


図 1 (1–9)

1: 黒は中央の線を取る。

9: 黒は三列、白は二列。

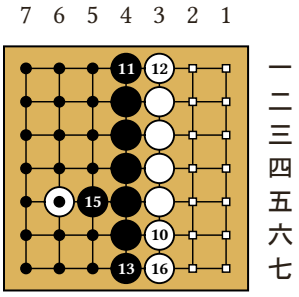


図 2 (10–18)

結果: 黒 0.5 目勝ち (黒 21, 白 20.5)

13: 手筋. 壁の最後の一手。

14: 悪手. 白の打ち込み。しかし二眼を作る余地はない。

15: 黒が急所に打つ。白石は死んでいる。

18: 黒は二十目と死石一つ。白は十四目とコミ六目半。

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 13cm, height: auto, margin: 0.5cm)

// The texts of gobanist follow the language of the text. The Japanese ones need a font that has them:
// MS Gothic comes with Windows; Noto Sans CJK JP is another one.
#set text(lang: "ja", font: ("Libertinus Serif", "MS Gothic"))

#gobanist-set(background: (fill: (color: rgb(220, 179, 92))),
  label: (at: (top, right), style: "japanese"))

#let kifu = sgf-read(path("kifu.sgf"))

#sgf-info(kifu, "GN"): #sgf-info(kifu, "PB")対#sgf-info(kifu, "PW")、コミ#sgf-info(kifu, "KM")。
#sgf-comment(kifu, 0)

#sgf-diagrams(kifu, step: 9, comments: true)
```

32. A game in Chinese

一盘短棋：黑方对白方，贴目 6.5。 七路棋盘上的对局，附有评注。

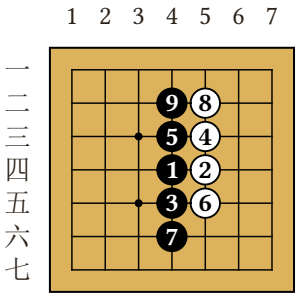


图 1 (1–9)

1: 黑棋占据中线。

9: 黑棋将有三列，白棋只有两列。

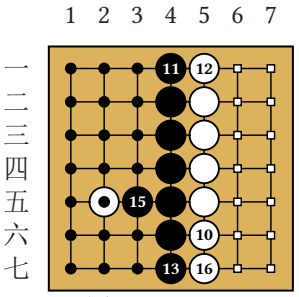


图 2 (10–18)

结果: 黑 胜 0.5 目 (黑 21, 白 20.5)

13: 手筋. 墙的最后一点。

14: 恶手. 白棋打入，但没有做两只眼的空间。

15: 黑棋占据要点，白子已死。

18: 黑棋：二十目加一个死子。白棋：十四目加六目半贴目。

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 13cm, height: auto, margin: 0.5cm)

// The Chinese texts need a font that has them: SimSun comes with Windows; Noto Serif CJK SC is another one.
#set text(lang: "zh", font: ("Libertinus Serif", "SimSun"))

#gobanist-set(background: (fill: (color: rgb(220, 179, 92))),
  label: (at: (top, left), style: "chinese"))

// qipu.sgf is in GB2312 (CA[GB2312]): sgf-read converts it.
#let qipu = sgf-read(path("qipu.sgf"))

#sgf-info(qipu, "GN"): #sgf-info(qipu, "PB")对#sgf-info(qipu, "PW")，贴目#sgf-info(qipu, "KM")。
#sgf-comment(qipu, 0)

#sgf-diagrams(qipu, step: 9, comments: true)
```

33. A game in Korean

짧은 대국: 흑 대 백, 덤 6.5. 7줄 바둑판에서 둔 대국과 해설.

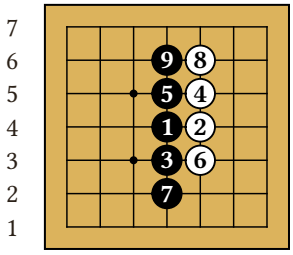


그림 1 (1–9)

1: 흑이 가운데 줄을 차지한다.

9: 흑은 세 줄, 백은 두 줄뿐이다.

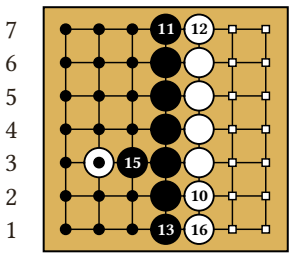


그림 2 (10–18)

결과: 흑 0.5 집 승 (흑 21, 백 20.5)

13: 맥점. 벽의 마지막 점.

14: 악수. 백이 침입하지만 두 눈을 만들 공간이 없다.

15: 흑이 급소를 차지한다. 백돌은 죽었다.

18: 흑: 20집과 사석 하나. 백: 14집과 덤 6.5집.

```
#import "@preview/gobanist:0.2.0": *
#set page(width: 13cm, height: auto, margin: 0.5cm)

// The Korean texts need a font that has them: Malgun Gothic comes with Windows; Noto Serif CJK KR is another
// one.
#set text(lang: "ko", font: ("Libertinus Serif", "Malgun Gothic"))

#gobanist-set(background: (fill: (color: rgb(220, 179, 92))), label: (at: (bottom, left)))

// gibo.sgf is in EUC-KR (CA[EUC-KR]): sgf-read converts it.
#let gibo = sgf-read(path("gibo.sgf"))

#sgf-info(gibo, "GN"): #sgf-info(gibo, "PB") 대 #sgf-info(gibo, "PW"), 덤 #sgf-info(gibo, "KM").
#sgf-comment(gibo, 0)

#sgf-diagrams(gibo, step: 9, comments: true)
```